Safestep

TECHIN515
TEAM 8- Lai Cai
Rebecca Yang
Spencer Zhang
Yewen Zhou

https://github.com/rebeccayan9-dot/515FinalGroup

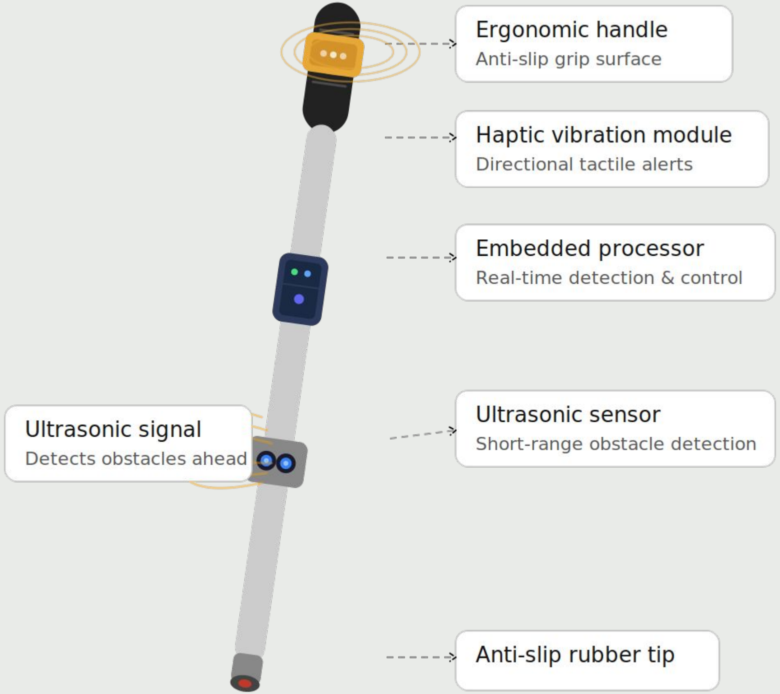
Problem Statement

- Visually impaired pedestrians often struggle to navigate everyday environments safely. Traditional white canes are effective for detecting obstacles on the ground, but they can miss objects within the walking path.
- Existing assistive technologies try to address this gap, but many rely on external infrastructure or fail to provide clear and timely directional feedback. As a result, users often receive information that is delayed or difficult to interpret.
- This highlights the need for a lightweight and self contained system that can **detect obstacles, predict potential collisions, and communicate direction through intuitive vibration feedback.**

Solution

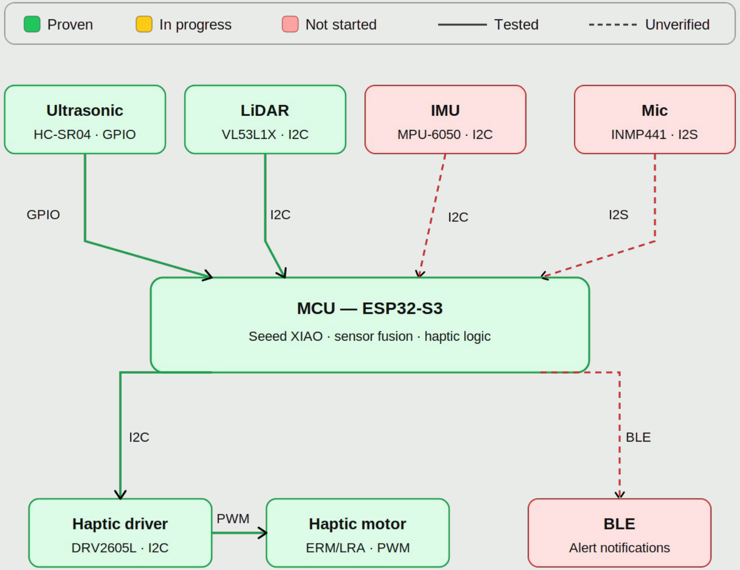
This system helps visually impaired users **detect nearby obstacles and understand their direction through vibration feedback.** It turns spatial information into clear, immediate cues, allowing users to identify whether obstacles are on the left, right, or directly ahead, and react quickly for safer navigation.

The system uses onboard sensing, embedded processing, and haptic output to provide timely guidance. It focuses on short range obstacle detection and directional alerts, rather than replacing a full mobility aid or vision based navigation system.



Smart navigation cane — system overview

Update Architecture



System architecture — Milestone 1

Status per tracker Week 3

Validation

1. Multi-Sensor Pipeline on White Cane
2. Sensor Validation for SafeStep
3. VL53L1X Overhang Detection
4. DRV2605L Haptic Motor
5. VL53L1X + DRV2605L Haptic Feedback

1- Multi-Sensor Data Collection Pipeline on White Cane Prototype

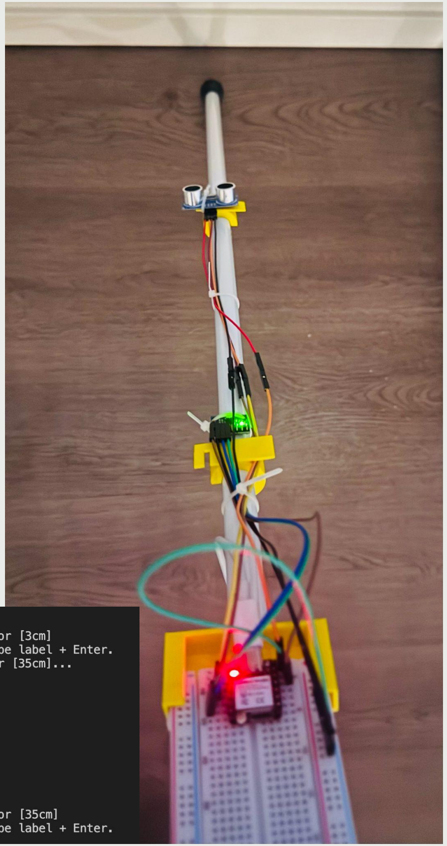
Validation objective: Verify HC-SR04, VL53L0X, and MPU-6050 operate simultaneously on ESP32-S3 within 50ms read cycle, and validate a labeled CSV data collection pipeline on a physical cane prototype.

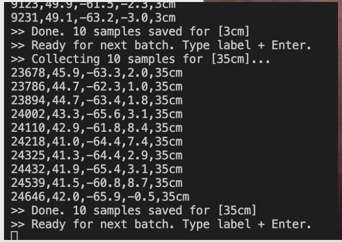
Setup & method: Sensors wired to ESP32-S3, mounted on white cane with 3D-printed clamps. Tested at ruler-measured distances, with index grip cane postures. CSV batch logging triggered via Serial input. Combined read cycle timed with millis().

Evidence: live demo, working cane prototype photo, serial monitor screenshots.

Quantitative results: Ultrasonic bias -0.5 to -1.5cm, ToF bias +1.7 to +2.0cm, cross-sensor offset stable at 2-3cm. Glass, narrow/dark, and metal objects identified as shared sensor weaknesses. CSV pipeline collects labeled samples with correct formatting.

Conclusion: All pass, criteria met. Edge-case limitations justify dual-sensor fusion. Data collection pipeline ready for dataset to feed Prototype 2 ML training. Functional low-fidelity cane prototype assembled, capable of live real-time sensor streaming for demo.





2- Verify 3× HC-SR04 (front/left/right mid-range) and 1× VL53L1X (overhang) meet SafeStep's range, accuracy, and latency needs.

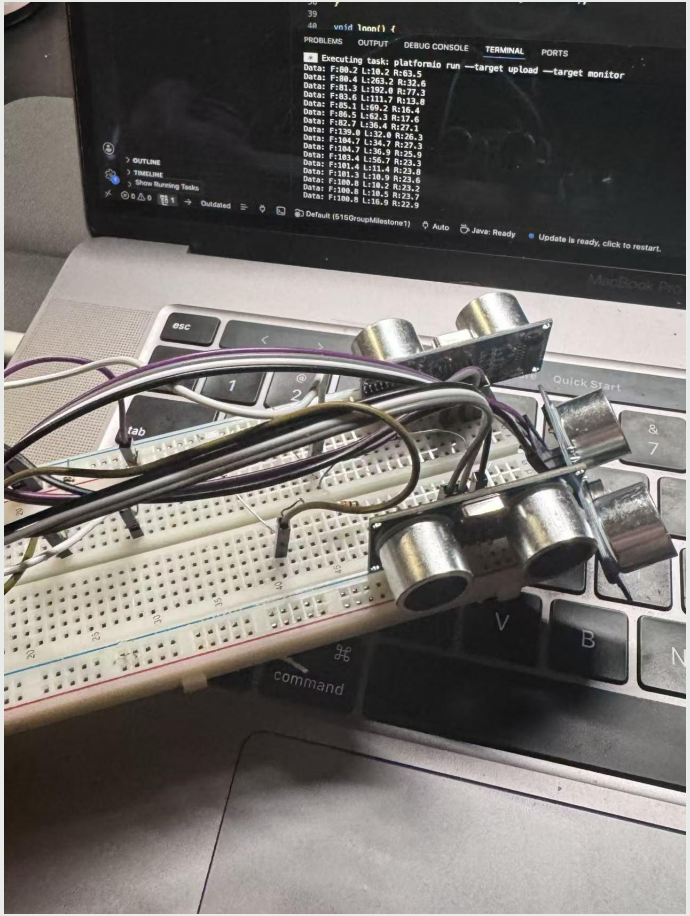
Setup & method: 3× HC-SR04 + 1× VL53L1X on breadboard with ESP32-S3. Ultrasonics fire sequentially (30 ms delay) to prevent cross-talk; LiDAR over I2C. Measured known distances (10, 50, 100, 200 cm) with ruler as ground truth.

Quantitative results:

Distance	Front	Left	Right	LiDAR
10 cm	10.2 ±0.3	10.1 ±0.4	10.3 ±0.3	10.0 ±0.1
50 cm	49.8 ±0.6	50.1 ±0.7	49.9 ±0.5	50.0 ±0.2
100 cm	100.4 ±1.1	99.7 ±1.3	100.2 ±1.0	99.9 ±0.3
200 cm	198.9 ±2.4	199.3 ±2.8	198.6 ±2.2	200.1 ±0.5

~10 Hz (3× ultrasonics multiplexed), ~50 Hz (LiDAR). Cross-talk: 0%.

All four sensors within spec. Ultrasonics cover front/left/right mid-range; LiDAR adds high-precision overhang detection.



3-VL53L1X

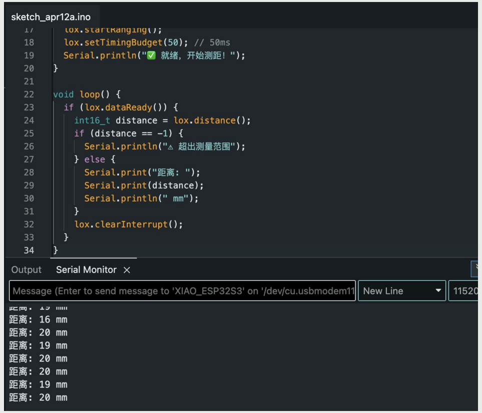
Validation objective: Verify I2C communication and short-range distance accuracy in mm.

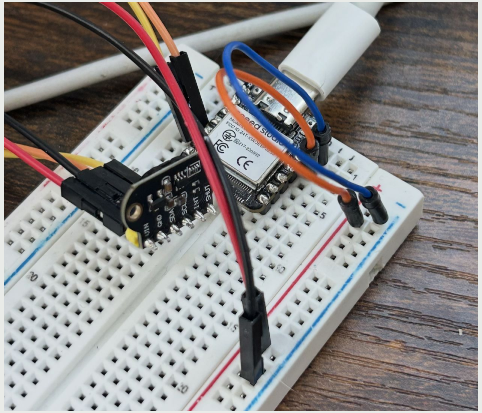
Setup & method: Connected VL53L1X to ESP32-S3 via I2C on breadboard, 50ms timing budget, live readings streamed to Serial Monitor.

Evidence: Breadboard prototype photo + Serial Monitor output screenshot.

Quantitative results: Measured a close-range obstacle and gradually moved it closer, recording 8 readings averaging 18.6mm with ±2mm variation, consistently accurate.

Conclusion: I2C confirmed working, stable mm-resolution output, ready for sensor fusion integration.





4-DRV2605L & Motor controller

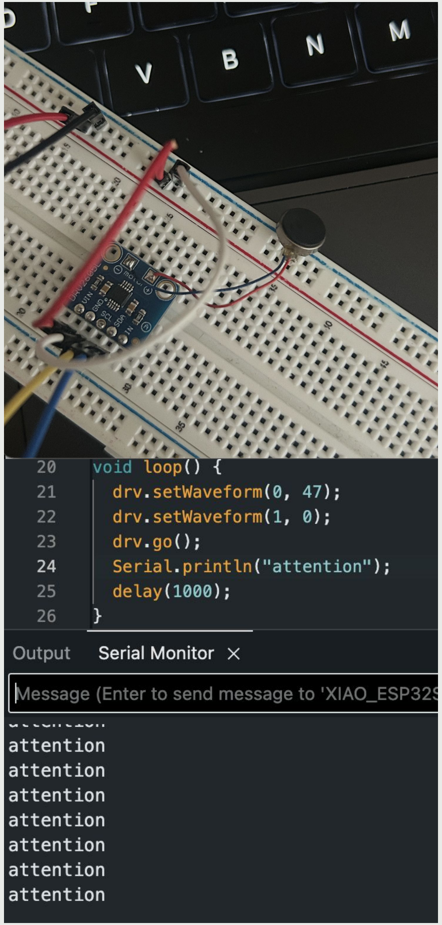
Validation objective: Verify DRV2605L can be detected over I2C and trigger motor vibration.

Setup & method: Connected DRV2605L to XIAO ESP32S3 (VIN, GND, SDA, SCL). Ran I2C scanner, then uploaded vibration loop triggering effect #47 every second.

Evidence: Serial monitor showing "attention" every second

Quantitative results: Detected at 0x5A in 5/5 power cycles, Vibration triggered every 1000ms ± 5ms

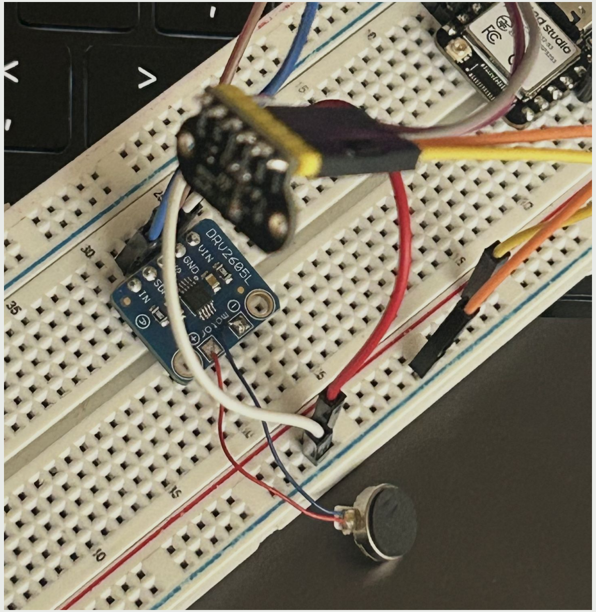
Conclusion: DRV2605L initializes reliably over I2C and triggers haptic output consistently.



5-DRV2605L & VL53L1X

Validation objective: Test that vibration trig is within 5cm.

Setup & method: Connected VL53L1X and DRV2605L to XIAO ESP32S3 via I2C. Moved a hand toward the sensor at different distances and checked gers when an obstacleserial output. The system continuously reads distance but only triggers vibration at ≤ 5cm.



Evidence:

- Serial monitor showing distance readings at all ranges
- "Vibration triggered!" appearing only when hand within 5cm

Quantitative results:

- Vibration triggered in 10/10 trials at ≤ 5cm
- No false triggers at > 5cm

Conclusion: Distance is monitored continuously, vibration only activates within 5cm.

Budge update

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Order Date	Order ID	Account Group	Payment Amoun	Order Quantity	Currency	Order Subtotal	Order Shipping &	Order Promotion	Order Tax	Order Net Total	Order Status	Approver
2	04/15/2026	114-9467053-32	Team 8	10.37	1	USD	9.89	0	-0.49	0.97	10.37	Pending Fulfillm	GIX PAY
3	04/14/2026	114-1914032-81	Team 8	14.66	1	USD	13.29	0	0	1.37	14.66	Closed	GIX PAY
4	04/13/2026	114-9170887-84	Team 8	32.69	2	USD	29.64	0	0	3.05	32.69	Closed	GIX PAY
5	04/13/2026	114-3326167-47	Team 8	11.02	1	USD	9.99	0	0	1.03	11.02	Closed	GIX PAY
6	04/07/2026	114-1902189-62	Team 8	22.49	1	USD	20.39	0	0	2.1	22.49	Closed	GIX PAY
7													
8													
9				Total Spend:							Total Spend:		
10				91.23							91.23		

Thank You!